Tathabbut at IslamicEval 2026: Evidence-Grounded Verification and Selective Canonical Correction

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Abstract

Islamic question-answering systems can produce fluent Arabic while altering Qur'anic wording, Hadith text, narrator chains, or source attribution. We present Tathabbut for two IslamicEval 2026 settings: fine-grained verification and canonical correction. For verification, **Canonical-Provenance Evidence Fusion (CPEF)** classifies *Ayah*, *matn*, *isnad*, and claimed-source segments by combining canonical retrieval with collection, narrator, response-level, contradiction, and exact-*matn* provenance evidence. CPEF achieves **0.9452 macro accuracy** on the official Test set. For correction, **Type-Aware Selective Canonical Retrieval (TASCR)** reranks authoritative Qur'anic and Hadith candidates using lexical, memory-based, contextual, and source evidence, and abstains when confidence is insufficient. TASCR achieves **86.29% Test accuracy**. A post-hoc Development diagnostic finds the gold correction for 319/327 (97.55%) recoverable cases in an expanded candidate pool, indicating that candidate discrimination and abstention are larger bottlenecks than retrieval recall. Neither submitted component invokes a generative LLM at inference time.

1 Introduction

Hallucination in religious question answering is not limited to obviously fabricated text. A generated response may preserve the style and broad semantics of an authentic quotation while changing one token, conflating narrations, assigning the wrong collection, or supplying an unsupported narrator chain. These errors call for fine-grained verification against explicit evidence rather than response-level plausibility alone. Related work on factuality has likewise emphasized atomic-fact evaluation (Min et al., 2023), claim-level reference checking (Hu et al., 2024), and multilingual hallucination-span detection (Vazquez et al., 2025; Hikal et al., 2025).

IslamicEval 2025 established a shared-task setting for hallucination in Qur'anic and Hadith content (Mubarak et al., 2025). Participating systems explored exact and normalized matching, fuzzy alignment, character-level retrieval, semantic reranking, and retrieval-augmented verification (Omayrah et al., 2025; Al Adel et al., 2025; ElKoumy et al., 2025). IslamicEval 2026 extends this line of work with fine-grained verification and correction tasks (Alharbi et al., 2026).

We address two complementary decision settings in IslamicEval 2026. Verification asks whether four annotated segment types—*Ayah*, *matn*, *isnad*, and claimed source—are correct or incorrect. Correction instead requires the exact canonical replacement for an erroneous *Ayah* or *matn*, while permitting abstention when a reliable correction cannot be identified. The two settings share evidence sources but require different decisions: verification asks whether the available evidence supports a claim, whereas correction must choose a specific canonical target and determine whether returning it is justified.

The systems share normalization, canonical grounding, source-aware evidence, and group-aware validation, while retaining task-specific decision mechanisms. Our contributions are: (1) CPEF, a relational verification framework combining textual authenticity with source and narrator provenance; (2) TASCR, an abstention-aware correction framework integrating hard-negative reranking, hallucination memory, contextual/source evidence, and type-specific routing; and (3) analysis of official submissions showing the value of treating source metadata as supporting evidence and of using conservative intervention and calibrated abstention.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1075–1080
October 28-29, 2026 ©2026 Association for Computational Linguistics

2 Tasks, Data, and Shared Processing

Subtask 2. The official score is the unweighted mean of the four segment-level accuracies,

$$\text{MacroAcc} = \frac{1}{4} \sum_{t \in \mathcal{T}} A_t, \tag{1}$$

where $\mathcal{T} = \{\text{Ayah, matn, isnad, source}\}$. Gold `N/A` rows are excluded. The released data contain 24,088 scored Training rows, 1,745 Development rows, and 10,375 Test rows. Test contains 3,539 *Ayah*, 3,076 *matn*, 465 *isnad*, and 3,295 claimed-source instances.

Subtask 3. The correction data contain 7,909 Training, 943 Development, and 1,998 Test instances. Training contains 3,703 *Ayahs* and 4,206 *matns*; Development contains 476 and 467; Test contains 944 and 1,054, respectively. A prediction is either the required canonical text or an abstention decision.

Canonical resources and text handling. Both systems use only the task-provided canonical resources. The Qur'an resource contains 6,236 address records (6,058 unique text strings after exact deduplication in the Subtask 3 pipeline). The Hadith resource contains 34,994 records; 31,811 have populated *matn*, yielding 30,940 unique *matns* after exact deduplication for correction retrieval. We maintain the authoritative canonical string separately from the representation used for search. Search normalization applies Unicode NFKC, removes *tatweel*, diacritics and Qur'anic marks where appropriate, normalizes whitespace and Arabic/Persian digits, strips punctuation/symbols, and folds conservative Arabic orthographic variants. Canonical output is always taken from the preserved resource text rather than reconstructed from its normalized form.

Grouping, cross-fitting, and leakage control. Segments from the same generated response or question can share wording, source strings, and local context. We therefore use group-aware splits whenever learned or memory-based evidence is estimated. The verification pipeline uses five-fold `GroupKFold` by `Response_ID`; the correction pipeline uses question-grouped splits for threshold calibration, hallucination-memory features, and confidence routing. During cross-fitting, each Training segment receives learned evidence only from a model fitted on the remaining response groups, and these out-of-fold predictions become features for later stages. This prevents a fragment from receiving learned or memory evidence derived from a correlated fragment in the same group.

3 Subtask 2: Canonical-Provenance Verification

3.1 Canonical and Collection-Conditioned Evidence

CPEF treats verification as agreement among several evidence sources rather than as four independent text-classification problems. For Qur'an, we index individual verses together with concatenations of up to six adjacent verses from the same surah, enabling partial and multi-verse quotations to retrieve an appropriate canonical span. A source parser normalizes surah aliases and verse numbers, allowing a strong textual match to be rejected when the claimed location is incompatible with the retrieved evidence.

For Hadith, *global retrieval* asks whether a *matn* resembles any canonical narration, whereas *collection-conditioned retrieval* asks whether comparable evidence exists within the claimed collection. Character n-gram TF–IDF with cosine similarity is complemented by fuzzy partial and token-set similarity. Source aliases are normalized, and "agreed upon" is mapped to al-Bukhari and Muslim. A quotation may therefore resemble an authentic narration globally while its specific attribution remains unsupported.

3.2 Type-Specific Verification and Provenance

Ayah and matn. Ayah evidence includes canonical similarity, containment, retrieval margin, and source-location agreement. Matn verification combines global retrieval with collection-conditioned support. The override thresholds are empirical rather than theoretically derived. They were frozen from response-grouped Training audits that prioritized precision over coverage, because a mistaken override can convert a correct decision into an unsupported religious claim. The submitted collection-conditioned rescue requires source confidence ≥ 0.70, no unsupported collection, partial similarity ≥ 0.80, token-set similarity ≥ 0.96, and cosine similarity ≥ 0.50. A response-consistency rule can also rescue a matn that contains at least three normalized words and is strictly shorter, under the rule's normalized representation, than another already-supported matn in the same response;

the pair must reach partial/token-set similarities of at least 0.94/0.92. Here, "short" is defined relative to the supporting matn rather than by a universal Hadith-length cutoff. Grouped Training audits selected 92 and 25 cases for these rules, respectively, with 100% precision in those audits.

Claimed source. Source verification evaluates the relation among the *quotation, canonical candidate, and claimed source*, rather than classifying the source string in isolation. For example, a quotation may retrieve an authentic narration in Muslim while the generated response attributes it to al-Bukhari; in that case, the wording can be authentic-like even though the source relation is unsupported. Qur'anic claims are checked analogously against retrieved verse locations. Hadith source evidence includes global and collection-restricted scores and margins, exact/near-exact matn evidence, number agreement, cross-fitted textual evidence, and relational source–matn memory. Compact learned source models contribute evidence, while conservative routing preserves high-confidence canonical matches and explicit contradictions.

Isnad. The five nearest canonical Hadith matns are retrieved using character-window TF–IDF. Narrator extraction is deliberately conservative rather than based on unrestricted named-entity detection: candidate spans are initiated only by recognized Arabic transmission cues and narration verbs used by the submitted extractor, terminated at reporting, Prophet-reference, or honorific markers, and capped at seven normalized tokens. Simple normalized variants account for common kunyah/name forms and selected shortened *ibn* chains. The resulting features include exact narrator-phrase matches, narrator-token recall, retrieval scores and margins, claimed-book agreement, book-conditioned similarity, and chain/text lengths. A character-level Linear SVM (2–6 grams, $C = 0.5$) supplies out-of-fold lexical evidence, while a 500-tree Random Forest (depth 5, minimum leaf 5, balanced classes) fuses lexical and narrator-provenance features. For claims tied to the six canonical books, the submitted system uses a 0.45 probability threshold; external or empty metadata retain the base decision unless canonical narrator evidence supports an override.

3.3 Directional Contradiction and Provenance Memory

The final system applies asymmetric overrides. Positive rescues require convergent authenticity evidence, whereas negative overrides require localized contradiction evidence. A rare token-insertion rule rejects a near-canonical Hadith only when similarity is high, one-to-three tokens are inserted, no canonical token is deleted, the inserted material is rare, and either a parenthetical gloss or a collection conflict is present. The rule selected seven grouped Training cases, all incorrect. A Qur'anic contradiction rule requires retrieval similarity ≥ 0.90, a parsed source that conflicts with the retrieved verse, and grouped $P(\text{correct}) \leq 0.15$; it selected 24 incorrect Training cases.

The final CPEF component is **exact-matn source-provenance memory**. For each exact normalized matn, the system records positive/negative parent counts, per-book support, and exact source-set counts. An incorrect source prediction is rescued only when every claimed collection has positive support for the same matn, contradictory exact-source evidence is absent, and the support is repeated (the exact source set occurs at least twice, or every claimed collection has at least five positive observations). Across grouped Training folds, the complete memory rule selects 619 cases at 0.9935 precision. This is relational rather than direct label memorization: the decision depends on the consistency of the matn–source association.

4 Subtask 3: Type-Aware Selective Canonical Correction

4.1 Retrieval, Reranking, and Hallucination Memory

TASCR formulates correction as selective ranking over the official inventories. Each inventory is indexed with character `char_wb` 3–5-gram TF–IDF and word 1–2-gram TF–IDF. For Ayahs, up to 20 candidates from each view are retained; for Hadith, up to 10 are retained from each view and then deduplicated. Candidate features include cosine scores and ranks, retriever agreement, score margins, full/partial/token-set fuzzy ratios, token Jaccard and coverage, and length features. A LightGBM reranker is trained with hard retrieved negatives: each available positive is paired with the ten strongest negative candidates rather than with random corpus items. A question-grouped 80/20 split calibrates the candidate-versus-abstention thresh-

old before the reranker is refit on the selected Training pairs.

The **hallucination memory** separately indexes labeled erroneous Training spans with character 3–5-gram TF–IDF. The ten closest hallucinations vote for their known corrections using squared-cosine weights. Features summarize the strongest support, vote share and margin, agreement count, support for abstention, and agreement with the current canonical candidate. Training-time memory features are computed out of fold with five question-grouped folds. When the memory provides sufficiently strong support, it may inject a canonical correction that was absent from the lexical candidate pool.

4.2 Context, Source Evidence, and Selective Routing

Candidate evidence is augmented with character- and word-level similarity to the original question, the local response context, and their concatenation. Source strings are normalized separately, and numerical references, Qur'anic addresses, and aliases of the six Hadith collections are compared with candidate metadata. These source signals remain soft evidence because the generated source itself may be hallucinated.

Development produced two complementary experts: a memory-aware lexical reranker and a context/source-aware reranker. A precursor uses the context/source expert for Ayahs and the memory expert for matns. TASCR adds a second-level Hadith confidence router that evaluates the distinct actions proposed by both experts together with an abstention action. Its 51 global/action-specific features include top and runner-up scores, margins to calibrated thresholds, lexical and memory evidence, context/source support, abstention states, expert agreement, and confidence gaps. The router is a regularized LightGBM model (180 trees, learning rate 0.03, depth 4, seven leaves) assessed with five-fold question-grouped cross-validation. Because routing improves matn correction but slightly reduces Ayah accuracy, the final policy is type-specific: Ayah $\rightarrow$ context/source expert; matn $\rightarrow$ confidence router.

4.3 Diagnostic Candidate Coverage

A post-hoc Development diagnostic expands retrieval across both official inventories and adds source-guided candidates. This diagnostic is not part of the submitted TASCR inference path. Among 327 recoverable Development examples, the gold correction appears in the expanded pool for 319 (97.55%): 216/219 Ayahs and 103/108 matns. Together with 616 Development cases whose correct action is abstention, the candidate-pool oracle is 935/943 (99.15%). The gap between candidate availability and actual accuracy shows that recall alone is insufficient: the system must still distinguish confusable canonical neighbors and determine when abstention is safer.

Subtask 2: official Test accuracy					
Approach	Ayah	Source	Isnad	Matn	Macro
CCMA	.9825	.9590	.9312	.9057	.9446
TECV	.9825	.9590	.9312	.9064	.9448
CPEF	**.9828**	**.9605**	.9312	**.9064**	**.9452**

Table 1: Official Test results for the three submitted Subtask 2 systems.

Subtask 3: official Test accuracy (%)			
Approach	Overall	Ayah	Matn
Type-Routed Hybrid	86.09	**84.53**	87.48
TASCR	**86.29**	**84.53**	**87.86**
Exact-Source Rescue	86.19	84.32	**87.86**

Table 2: Official Test results for the three submitted Subtask 3 systems.

5 Results and Analysis

Subtask 2. CPEF is the strongest submitted verification system. It yields 3,478/3,539 correct Ayahs, 3,165/3,295 claimed sources, 433/465 isnads, and 2,788/3,076 matns. Relative to CCMA, TECV changes four matn decisions and produces a net gain of two correct matns. CPEF then changes six decisions relative to TECV—one Ayah rejection and five source rescues—and all six are beneficial on Test. Although the number of changed decisions is too small to support a significance claim, their outcomes are consistent with the intended high-precision behavior of the final contradiction/provenance layer.

The Development-to-Test gap is strongly type dependent. CPEF moves from 0.9884 Development macro accuracy to 0.9452 on Test. Ayah changes from 0.9928 to 0.9828 and claimed source from 0.9814 to 0.9605, whereas matn falls from 0.9796 to 0.9064 and isnad from 1.0000 to 0.9312. The Development isnad estimate is particularly fragile because it contains only 30 scored chains, compared with 465 at Test time. The remaining matn errors are consistent with partial quotations, re-

ordered clauses, and collection-specific variants, while isnad remains sensitive to narrator aliases, abbreviated chains, and multiple transmission paths.

Subtask 3. TASCR is the strongest submitted correction system, with 1,724/1,998 correct predictions (86.29%). Relative to Type-Routed Hybrid Retrieval, it adds four correct Hadith predictions while preserving Ayah accuracy. The Exact-Source Conservative Rescue run is informative: it improves one Development decision but reduces Test Ayah accuracy from 84.53% to 84.32% while leaving matn accuracy unchanged. This result illustrates that source parseability and specificity are not equivalent to source reliability; a precise citation can itself be hallucinated.

The candidate-coverage diagnostic further separates retrieval from selection. Only eight of 327 recoverable Development cases lack the gold correction in the expanded pool. Nevertheless, TASCR correctly recovers 216/327 recoverable quotations while correctly abstaining on 582/616 gold-abstention cases. This conservative profile reduces false canonical replacements but leaves substantial missed-correction error. Additional Development experiments reinforce the same interpretation: zero-shot multilingual E5 dense fusion reaches 83.78%, and an expanded-retrieval rescue reaches 84.52% but changes 14 abstentions at only 35.7% rescue precision. In this exact-match setting, broader semantic relatedness or additional candidate recall cannot substitute for calibrated candidate discrimination.

Shared lessons. Across both subtasks, canonical evidence is necessary but not sufficient. Source metadata should be treated as evidence that may agree or conflict with the text, not as an infallible key. Hadith requires more conservative handling than Qur'an because its inventory is larger and contains more overlapping and collection-dependent wording. Group-aware estimation is also important because related fragments can leak response/question structure across folds. Finally, the decision mechanism is task-specific: verification benefits from relational contradiction/provenance rules, whereas correction requires calibrated abstention after obtaining a high-recall candidate set.

6 Limitations

Development data were inspected repeatedly during competition development, so progression numbers should be interpreted as model-selection evidence rather than as unbiased generalization estimates. The verification systems are sequential rather than factorial ablations, and the high-precision rule audits do not establish causal component effects. The rule thresholds and evidence combinations were developed for the IslamicEval resources; applying CPEF to other Hadith collections, editions, languages, or hallucination datasets would likely require recalibration rather than direct reuse. Exact-matn provenance memory cannot support unseen matns, and differences in wording, numbering, or edition may separate authentic evidence from the supplied canonical representation. For correction, the 97.55% candidate coverage is a post-hoc diagnostic over an expanded pool rather than part of the submitted inference path. Both systems rely primarily on lexical, contextual, and provenance proxies instead of explicit models of deep semantic entailment or full transmission-history graphs.

7 Ethical Considerations

Religious verification and correction are high-sensitivity applications. A false positive in verification can legitimize unsupported content, while a false correction can replace one hallucination with a different authentic quotation and thereby create a new attribution error. The systems should be viewed as research aids rather than substitutes for qualified Islamic scholarship. Practical use should expose retrieved canonical evidence, source information, and confidence or abstention decisions to human reviewers.

Generative AI tools were used to assist with manuscript organization, drafting, and language editing. The author reviewed the final manuscript and is responsible for the accuracy of the reported methods, experimental results, and citations.

8 Conclusion

Tathabbut unifies IslamicEval 2026 fine-grained verification and canonical correction. CPEF combines canonical, relational, narrator, contradiction, and provenance evidence, achieving 0.9452 Test macro accuracy. TASCR combines hard-negative reranking, hallucination memory, contextual/source evidence, type-aware routing, and explicit abstention, achieving 86.29% Test accuracy. Together, the systems support provenance-aware, conservative use of canonical retrieval.

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